Shortlisted Bibliography

PAT-scan Literature — User-Curated Stack (Stack 3)

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Eight papers enriched via OpenAlex from a user-curated tree of three categories: Mathematical Foundations, Statics-based Methods, Vibrations-based Methods.

Summary

Category	Count	Year range
Mathematical Foundations	1	1994
Statics-based Methods	5	2003–2022
Vibrations-based Methods	2	2007–2022
Total	**8**	**1994–2022**

1 Mathematical Foundations

[**1**] Raghavan, K. and Yagle, A. E.. *Forward and inverse problems in elasticity imaging of soft tissues.* IEEE Transactions on Nuclear Science, 1994. doi:10.1109/23.322961

University of Michigan–Ann Arbor. Cited by 99. Foundational formulation casting the elasticity inverse problem as a linear system via finite differences, with adaptive piston deformations. Establishes the consistency between forward and inverse problems exploited by all later work in this stack.

2 Statics-based Methods

[**2**] Konofagou, E. E. and Harrigan, T. P.. *Palpation tomography — a new technique for modulus estimation in elastography.* IEEE Ultrasonics Symposium Proceedings, 2003/2004. doi:10.1109/ultsym.2003.129348

Columbia University; Exponent (United States). Uses multiple smaller quasi-static load cases (inspired by clinical palpation) instead of one whole-boundary load, increasing the measurement-to-parameter ratio and reducing noise sensitivity in modulus reconstruction.

[**3**] Liu, Y., Sun, L., and Wang, G.. *Tomography-based 3-D anisotropic elastography using boundary measurements*. IEEE Transactions on Medical Imaging, 2005. doi:10.1109/tmi.2005.857232

University of Iowa; University of Iowa Hospitals and Clinics. Cited by 41. Anisotropic elastography framework using static displacements on the entire boundary plus partial surface force data, with adjoint-method gradients accelerating the optimization-based reconstruction.

[**4**] Olson, L. G. and Throne, R. D.. *An inverse problem approach to stiffness mapping for early detection of breast cancer*. Inverse Problems in Science and Engineering, 2012. doi:10.1080/17415977.2012.700710

Rose–Hulman Institute of Technology. Cited by 13. 3D finite-element reformulation of the boundary force/deflection inverse problem, with a genetic algorithm identifying interior tissue moduli. Detects 1 cm tumours at 23 dB SNR.

[**5**] Mei, Y., Wang, S., Shen, X., Rabke, S., and Goenezen, S.. *Mechanics based tomography: a preliminary feasibility study*. Sensors (MDPI), 2017. doi:10.3390/s17051075

Texas A&M University. Cited by 19. Open access (gold). Coins the term *Mechanics Based Tomography* (MBT). Uses force sensors plus DIC-measured boundary displacements — no interior measurements — to reconstruct shear-modulus maps of stiff inclusions in a soft background.

[**6**] Zhang, E., Dao, M., Karniadakis, G. E., and Suresh, S.. *Analyses of internal structures and defects in materials using physics-informed neural networks*. Science Advances, 2022. doi:10.1126/sciadv.abk0644

Brown University; MIT; Nanyang Technological University. Cited by 344. Open access (gold). PINN framework for identifying unknown geometry and material parameters with a differentiable, trainable parameterization. Validated across linear elasticity, hyperelasticity, and plasticity — recovers void/inclusion shape, size, location, and modulus.

3 Vibrations-based Methods

[**7**] Peters, A.. *Digital image elasto-tomography: mechanical property reconstruction from surface measured displacement data*. Ph.D. thesis, University of Canterbury, 2007. doi:10.26021/2670

University of Canterbury. Open access (green). Introduces Digital Image Elasto-Tomography (DIET): steady-state surface motion captured by a camera array drives a finite-element-based stiffness reconstruction, combining deterministic and stochastic algorithms. Targets breast-cancer screening as a low-cost, high-contrast alternative to mammography.

[**8**] Feng, B. T., Ogren, A. C., Daraio, C., and Bouman, K. L.. *Visual vibration tomography: estimating interior material properties from monocular video*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022. doi:10.1109/cvpr52688.2022.01575

California Institute of Technology. Cited by 12. Open access (green). Estimates spatially-varying Young's modulus and density throughout a 3D object from a single monocular video, by extracting image-space modes from sub-pixel surface vibration and inverting to material properties. Validated on simulated and real videos.

Generated by the Stack 3 cap-stone skill (`/identify-papers-user`) over the curated tree at `paper-stack-three-test-two/shortlisted/`. Source: OpenAlex (resolution mode: search) on May 31, 2026.